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APPLIED MACHINE LEARNING – COMPREHENSIVE MCQ BANK
Topics: Unsupervised Learning | Neural Networks | Image Processing |
        Convolutional Neural Networks | Transfer Learning | NLP
Total : 150 Questions
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SECTION 1: UNSUPERVISED LEARNING
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- Q1. In unsupervised learning, what is the primary objective of clustering?
- A. To predict a target label for each data point
 - B. To group data points such that points within the same group are more similar to each other than to those in other groups
 - C. To reduce the number of features in the dataset
 - D. To find the decision boundary that best separates two classes
- Q2. Which of the following is a key distinction between supervised and unsupervised clustering?
- A. Clustering requires labelled training data while classification does not
 - B. Clustering discovers structure from unlabelled data without predefined class labels
 - C. Clustering always produces a fixed number of groups equal to the number of features
 - D. Clustering can only be applied to numeric features, not categorical ones
- Q3. A data scientist applies a clustering algorithm to customer purchase data and obtains clusters. Which metric is appropriate for evaluating cluster quality WITHOUT ground-truth labels?
- A. Accuracy
 - B. F1-Score
 - C. Silhouette Score
 - D. Adjusted Rand Index
- Q4. The silhouette coefficient for a data point is computed as $(b - a) / \max(a, b)$, where 'a' is the mean intra-cluster distance and 'b' is the mean nearest-cluster distance. A value close to +1 indicates:
- A. The point is on the boundary between two clusters
 - B. The point is far from its own cluster and close to neighbouring clusters
 - C. The point is well-matched to its own cluster and poorly matched to neighbouring clusters
 - D. The clustering algorithm has not converged
- Q5. In density-based clustering, clusters are defined as regions of:
- A. Minimum variance in feature values
 - B. High density separated by regions of low density
 - C. Equal Euclidean distances from a centroid
 - D. Maximum inertia within a fixed radius
- Q6. In the K-Means algorithm, which of the following best describes the assignment step?
- A. Each centroid is moved to the mean of all data points
 - B. Each data point is assigned to the centroid with the smallest Euclidean distance
 - C. New centroids are randomly sampled from the data
 - D. Clusters with zero members are split into two sub-clusters
- Q7. K-Means minimises which objective function?

- A. Sum of pairwise distances between all data points
 - B. Maximum distance from any point to its assigned centroid
 - C. Within-cluster sum of squared distances (inertia)
 - D. Between-cluster variance divided by within-cluster variance
- Q8. A major limitation of the standard K-Means initialisation is:
- A. It requires the data to be normalised before clustering
 - B. It can converge to local optima depending on the random starting centroids
 - C. It cannot handle numeric features
 - D. It always produces the same result regardless of random state
- Q9. K-Means++ improves upon standard K-Means by:
- A. Automatically determining the optimal number of clusters
 - B. Using a probabilistic initialisation that spreads initial centroids further apart
 - C. Replacing Euclidean distance with cosine similarity
 - D. Running K-Means on a random subset of the data to speed up initialisation
- Q10. In K-Means++, the probability that a data point x is chosen as the next centroid is proportional to:
- A. The inverse of its distance to the nearest existing centroid
 - B. The squared distance from x to the nearest existing centroid
 - C. The rank of x by distance to the dataset mean
 - D. A uniform probability independent of any existing centroids
- Q11. K-Means assumes that clusters are:
- A. Arbitrarily shaped and of arbitrary density
 - B. Convex, isotropic (spherical), and of similar size
 - C. Separated by linear decision boundaries
 - D. Hierarchically nested within each other
- Q12. What does the Elbow Method plot on its x-axis and y-axis respectively?
- A. Number of clusters (k) on the x-axis; Within-cluster sum of squares (inertia) on the y-axis
 - B. Inertia on the x-axis; Number of clusters (k) on the y-axis
 - C. Silhouette score on the x-axis; Number of clusters on the y-axis
 - D. Number of features on the x-axis; Cluster variance on the y-axis
- Q13. In the Elbow Method, the 'elbow point' is identified as:
- A. The value of k where inertia reaches zero
 - B. The value of k where adding another cluster yields a diminishing reduction in inertia
 - C. The value of k where inertia first exceeds the silhouette score
 - D. The smallest value of k that produces non-negative inertia
- Q14. A drawback of relying solely on the Elbow Method for selecting k is:
- A. It cannot be applied when inertia is non-negative
 - B. The elbow point is often ambiguous and subjective, especially in real-world data
 - C. It requires ground-truth labels to compute inertia correctly
 - D. It tends to overestimate the optimal number of clusters by a factor of 2
- Q15. In DBSCAN, a 'core point' is defined as a data point that:
- A. Is the closest point to the centroid of any cluster
 - B. Has at least MinPts neighbours within its epsilon-neighbourhood
 - C. Has the maximum density among all points in its cluster
 - D. Is equidistant from exactly two cluster centroids
- Q16. In DBSCAN, data points that are not core points and not reachable from any core point are labelled as:
- A. Border points
 - B. Centroids
 - C. Noise points (outliers)
 - D. Satellite points

Q17. Which advantage does DBSCAN have over K-Means?

- A. It is computationally faster on high-dimensional data
- B. It can discover clusters of arbitrary shape and naturally identifies outliers
- C. It always requires a smaller number of hyperparameters
- D. It guarantees convergence to the global optimum

Q18. DBSCAN struggles with datasets that have:

- A. Fewer than 100 data points
- B. Clusters of uniformly varying density across the dataset
- C. Clusters of significantly varying density, because a single epsilon cannot suit all clusters
- D. Features measured on the same scale

Q19. HDBSCAN improves upon DBSCAN primarily by:

- A. Using cosine distance instead of Euclidean distance
- B. Automatically selecting the epsilon parameter via grid search
- C. Building a cluster hierarchy and extracting stable clusters across varying density levels
- D. Replacing the MinPts parameter with an adaptive kernel bandwidth

Q20. The Adjusted Rand Index (ARI) compares:

- A. The within-cluster variance to the total data variance
- B. The clustering output to a ground-truth label assignment, corrected for chance
- C. The entropy of cluster assignments to the entropy of the true labels
- D. The silhouette score of two different clustering algorithms

Q21. An ARI value of 0 indicates:

- A. Perfect agreement between the clustering and the ground truth
- B. The clustering is no better than a random label assignment
- C. The clustering is completely wrong (inverse of the ground truth)
- D. The number of clusters equals the number of classes in the ground truth

Q22. Normalised Mutual Information (NMI) measures:

- A. The average distance between cluster centroids normalised by the dataset diameter
- B. The shared information between the predicted cluster labels and the true labels, normalised to the range [0, 1]
- C. The ratio of inter-cluster variance to intra-cluster variance
- D. The proportion of data points correctly assigned to their true clusters

Q23. In agglomerative hierarchical clustering, the algorithm starts by:
[Easy]

- A. Assigning all data points to a single cluster and recursively splitting it
- B. Treating each data point as its own cluster and iteratively merging the closest pair
- C. Randomly assigning data points to k clusters and re-assigning until convergence
- D. Computing the density of each point and expanding clusters from core points

Q24. Which linkage criterion in hierarchical clustering is most susceptible to the 'chaining' effect where elongated clusters form?

- A. Complete linkage
- B. Ward linkage
- C. Single linkage
- D. Average linkage

Q25. A dendrogram in hierarchical clustering represents:

- A. The 2D projection of high-dimensional cluster centroids
- B. A tree diagram showing the order and distances at which clusters are merged

- C. A heatmap of pairwise Euclidean distances between all data points
- D. The inertia curve plotted against the number of clusters

Q26. Principal Component Analysis (PCA) finds principal components by:

- A. Randomly projecting data into lower dimensions and selecting the best projection by accuracy
- B. Identifying directions of maximum variance in the data via eigenvalue decomposition of the covariance matrix
- C. Clustering data points and using cluster centroids as the reduced representation
- D. Applying a kernel function to map data to a higher-dimensional space

Q27. If the first two principal components of a dataset explain 85% of the total variance, the remaining components account for:

- A. Noise only, and should always be discarded
- B. 15% of the variance, which may still contain meaningful information depending on the application
- C. 85% of the variance, because PCA components are symmetric
- D. Exactly zero variance, since PCA is lossless

SECTION 2: NEURAL NETWORKS

Q28. A biological neural network inspires artificial neural networks because:

- A. Both process information in identical mathematical ways
- B. Both use connected units that receive signals, process them, and pass an output signal forward
- C. Both have identical learning rates and activation thresholds
- D. Biological neurons operate in discrete time steps matching gradient descent epochs

Q29. In the context of neural networks, 'representation learning' refers to:

- A. Learning to represent the training loss in a closed-form equation
- B. The network's ability to automatically learn useful feature representations from raw data
- C. Storing the training data in the network weights for later retrieval
- D. Translating one programming language representation to another

Q30. The 'universal approximation theorem' states that a feedforward network with a single hidden layer containing a sufficient number of neurons can:

- A. Approximate any convex function exactly with zero error
- B. Approximate any continuous function on a compact subset of real numbers to arbitrary accuracy
- C. Learn any function in polynomial time regardless of dataset size
- D. Generalise perfectly to unseen data if trained long enough

Q31. An artificial neuron computes its output by:

- A. Randomly selecting one of its inputs and passing it to the next layer
- B. Computing the weighted sum of inputs, adding a bias, and passing the result through an activation function
- C. Dividing the input vector by the number of neurons in the previous layer
- D. Computing the maximum of all input values and squaring it

Q32. The bias term in an artificial neuron serves to:

- A. Penalise large weight values and prevent overfitting
- B. Shift the activation function, allowing the neuron to fire even when all inputs are zero
- C. Normalise the input values to the range [0, 1]
- D. Set the learning rate during backpropagation

- Q33. The single-layer perceptron can only solve classification problems that are:
- A. Non-linearly separable
 - B. Linearly separable
 - C. Multi-class with more than two categories
 - D. High-dimensional with at least 100 features
- Q34. The XOR problem revealed a fundamental limitation of the single-layer perceptron. Which architectural extension resolved this limitation?
- A. Adding more input neurons
 - B. Adding one or more hidden layers to create a Multi-Layer Perceptron
 - C. Replacing the step activation function with a linear function
 - D. Increasing the learning rate to allow faster convergence
- Q35. In a fully connected (dense) layer with 128 input neurons and 64 output neurons, how many trainable weight parameters exist (excluding biases)?
- A. $128 + 64 = 192$
 - B. $128 \times 64 = 8,192$
 - C. $64 \times 64 = 4,096$
 - D. $128 / 64 = 2$
- Q36. In a feedforward neural network, information flows:
- A. Bidirectionally between all layers during both training and inference
 - B. From the input layer through hidden layers to the output layer without cycles
 - C. From the output layer back to the input layer during inference
 - D. Only between adjacent hidden layers, never to or from the output layer
- Q37. What is the primary purpose of hidden layers in a neural network?
- A. To store training data for retrieval during inference
 - B. To learn intermediate feature representations that are not directly observable in the raw input
 - C. To compute the loss function before the output layer
 - D. To ensure the output layer receives data in sorted order
- Q38. The ReLU (Rectified Linear Unit) activation function is defined as:
- A. $f(z) = 1 / (1 + e^{-z})$
 - B. $f(z) = \max(0, z)$
 - C. $f(z) = (e^z - e^{-z}) / (e^z + e^{-z})$
 - D. $f(z) = z$ for all z
- Q39. The 'dying ReLU' problem occurs when:
- A. ReLU outputs are too large, causing exploding gradients
 - B. Neurons receive consistently negative pre-activation values and permanently output zero, causing their gradient to be zero
 - C. ReLU is applied to the output layer in a regression task
 - D. The learning rate is set too low, making ReLU gradients vanish
- Q40. Leaky ReLU addresses the dying ReLU problem by:
- A. Clipping large positive values to a maximum threshold
 - B. Allowing a small non-zero gradient (a small slope α) for negative inputs
 - C. Replacing the zero region with a sinusoidal function
 - D. Normalising the output of ReLU to zero mean
- Q41. The softmax activation function is most appropriate for the output layer of a network performing:
- A. Binary classification with a single output neuron
 - B. Multi-class classification where exactly one class is true
 - C. Regression to a continuous target value
 - D. Dimensionality reduction in the bottleneck layer of an autoencoder
- Q42. Which property of the sigmoid activation function causes vanishing gradients in deep networks?

- A. It outputs values in the range (0, 1), but its derivative approaches zero for large positive or negative inputs
- B. It is non-differentiable at the origin
- C. It produces strictly negative outputs for negative inputs
- D. Its outputs are always greater than 1 for positive inputs

Q43. The tanh activation function differs from sigmoid in that:

- A. It is not differentiable anywhere
- B. Its outputs are zero-centred, ranging from -1 to 1, which reduces the zig-zagging of gradient updates
- C. It is only used in the output layer of networks
- D. It has a constant gradient of 1 for all inputs

Q44. During forward propagation, the pre-activation value z at a neuron is computed as:

- A. $z = \text{activation}(w \cdot x)$
- B. $z = w \cdot x + b$, where w are weights, x are inputs, and b is the bias
- C. $z = (w \cdot x) / ||w||$
- D. $z = \text{softmax}(x) + b$

Q45. In a 3-layer network (input, one hidden, output), forward propagation is performed:

- A. From the output layer to the input layer
- B. From the input layer through the hidden layer to the output layer
- C. Simultaneously at all layers in parallel
- D. Only at the hidden layer; input and output layers perform no computation

Q46. Why must activation functions be non-linear for deep networks to be expressive?

- A. Without non-linearity, weights cannot be updated by gradient descent
- B. Without non-linearity, a stack of linear layers collapses to a single linear transformation, removing the benefit of depth
- C. Without non-linearity, the network cannot represent negative values
- D. Without non-linearity, the output layer cannot use softmax

Q47. Which loss function is most appropriate for a binary classification problem where the model outputs probabilities?

- A. Mean Squared Error
- B. Binary Cross-Entropy
- C. Hinge Loss
- D. Mean Absolute Error

Q48. Binary cross-entropy loss $L = -[y \log(p) + (1-y) \log(1-p)]$ assigns a very large loss when:

- A. The predicted probability p is close to the true label y
- B. The predicted probability p is near 0.5 for any label
- C. The predicted probability p is far from the true label y (e.g., predicting 0.01 when $y=1$)
- D. The learning rate is too large

Q49. For a multi-class classification task with 10 classes, the appropriate loss function and final activation are:

- A. Binary cross-entropy and sigmoid
- B. Categorical cross-entropy and softmax
- C. Mean squared error and ReLU
- D. Hinge loss and tanh

Q50. Using Mean Squared Error as the loss function for a binary classification task is generally discouraged because:

- A. MSE cannot be differentiated with respect to the network weights
- B. MSE treats the problem as regression, which may cause slow learning when the output probabilities are far from 0 or 1 due to saturating gradients
- C. MSE is not defined for probability values
- D. MSE always converges to a local minimum rather than the global minimum

Q51. Backpropagation computes gradients by:

- A. Numerically perturbing each weight and measuring the change in loss
- B. Applying the chain rule of calculus to propagate the loss gradient from the output layer back through each layer
- C. Randomly assigning gradient values and selecting the ones that reduce loss
- D. Computing the second derivative of the loss to find the curvature

Q52. The vanishing gradient problem in deep networks means that:

- A. Gradients in the output layer grow too large, causing weight overflow
- B. Gradients become exponentially small as they are backpropagated to earlier layers, causing slow or no learning in those layers
- C. The gradient of the loss with respect to biases is always zero
- D. The learning rate decreases automatically during training

Q53. The exploding gradient problem is most commonly addressed by:

- A. Using the sigmoid activation in all hidden layers
- B. Gradient clipping, which rescales gradients if their norm exceeds a threshold
- C. Increasing the batch size to average out extreme gradients
- D. Removing the bias term from all layers

Q54. During backpropagation, the gradient of the loss with respect to a weight in layer L requires:

- A. Only the output of layer L and the learning rate
- B. The chain-rule product of gradients from all layers after L through to the loss
- C. A separate forward pass for each weight
- D. Computing the Hessian matrix of the loss function

Q55. In stochastic gradient descent (SGD), the gradient is estimated using:

- A. The entire training dataset
- B. A single randomly chosen training example per update
- C. The validation set to avoid overfitting
- D. A fixed subset of the training data selected once before training

Q56. Mini-batch gradient descent balances the trade-offs of SGD and full-batch GD by:

- A. Computing the exact gradient of the loss with reduced memory cost
- B. Using a small random subset (batch) of the data to compute an approximate gradient each update
- C. Updating all weights in parallel using a GPU-optimised exact gradient
- D. Decreasing the learning rate proportionally to the batch size

Q57. The Adam optimiser combines which two techniques?

- A. L1 regularisation and L2 regularisation
- B. Momentum (first moment estimate) and adaptive learning rates (second moment estimate)
- C. Dropout and batch normalisation
- D. Gradient clipping and weight decay

Q58. In Adam, the learning rate is adapted per-parameter. This means that parameters updated frequently receive:

- A. Larger effective learning rates than infrequently updated parameters
- B. Smaller effective learning rates than infrequently updated parameters
- C. Identical learning rates regardless of update frequency
- D. Learning rates that are set to zero after a fixed number of updates

Q59. Learning rate scheduling refers to:

- A. Fixing the learning rate at a constant value for the entire training process
- B. Systematically adjusting the learning rate during training, such as decaying it over epochs
- C. Using a different learning rate for each layer in the network

D. Choosing the learning rate based on validation accuracy rather than training loss

Q60. An epoch in neural network training refers to:

- A. One update of the model weights using a single batch
- B. One complete pass through the entire training dataset
- C. The number of layers in the network
- D. The number of neurons in the largest hidden layer

Q61. Early stopping as a training strategy involves:

- A. Stopping training when the training loss reaches zero
- B. Halting training when the validation loss stops improving to prevent overfitting
- C. Reducing the number of neurons after each epoch
- D. Freezing weights in the first half of the network after a fixed number of epochs

Q62. During training, the model is evaluated on a validation set (not the test set) because:

- A. The validation set is always larger than the test set
- B. Hyperparameter tuning on the validation set prevents data leakage from the test set, which must remain unseen until final evaluation
- C. The test set does not contain ground-truth labels
- D. Validation set loss is always lower than training set loss

Q63. A model that performs well on training data but poorly on unseen test data is said to be:

- A. Underfitting
- B. Overfitting
- C. Well-regularised
- D. Converged

Q64. Underfitting can often be addressed by:

- A. Increasing regularisation strength
- B. Reducing the number of training epochs
- C. Increasing model capacity (adding more layers or neurons) or training longer
- D. Adding dropout to each hidden layer

Q65. The bias-variance tradeoff in neural networks implies that:

- A. Increasing model complexity always reduces both bias and variance simultaneously
- B. Reducing bias (by increasing capacity) typically increases variance, and vice versa; the goal is to find an optimal balance
- C. Bias and variance are independent properties that can be optimised separately
- D. A model with zero bias and zero variance is always achievable with sufficient training data

Q66. L2 regularisation (weight decay) adds which term to the loss function?

- A. $\lambda \sum |w|$ – the sum of absolute weight values
- B. $\lambda \sum w^2$ – the sum of squared weight values
- C. $\lambda \max(w)$ – the maximum weight value
- D. $\lambda \sum(w)$ – the sum of all weight values

Q67. L1 regularisation tends to produce sparse weight vectors because:

- A. Its gradient is always proportional to the weight magnitude
- B. Its gradient is constant in magnitude (sign of w), driving small weights to exactly zero
- C. It squares the weights, making small weights approach zero exponentially
- D. It clips gradient updates below a threshold

Q68. During training, Dropout randomly sets a fraction of neurons' outputs to zero. The primary effect is:

- A. Reducing the computational cost of forward propagation
- B. Preventing co-adaptation of neurons, which forces the network to learn redundant representations
- C. Increasing the gradient magnitude in backpropagation
- D. Ensuring the output layer always receives the full signal

Q69. At inference (test) time, Dropout is typically handled by:

- A. Applying dropout with the same rate as during training
- B. Disabling dropout and scaling the remaining neuron outputs by $(1 - \text{dropout rate})$
- C. Applying dropout only to the input layer
- D. Running the network 100 times with dropout enabled and averaging the predictions

Q70. Batch Normalisation normalises the activations within a mini-batch to have zero mean and unit variance, which primarily helps by:

- A. Increasing the learning rate exponentially during training
- B. Reducing internal covariate shift, allowing higher learning rates and stabilising training
- C. Replacing the need for biases in all layers
- D. Eliminating the need for regularisation altogether

Q71. Initialising all weights to zero in a neural network is problematic because:

- A. Zero weights cause the loss function to be undefined at the start of training
- B. All neurons in a layer compute identical gradients and learn identical features (symmetry problem), preventing the network from learning diverse representations
- C. Zero weights prevent forward propagation from producing any output
- D. Zero-initialised weights are too small for the activation functions to produce non-zero outputs

Q72. Xavier (Glorot) initialisation sets the initial weights to scale with:

- A. The square root of the learning rate
- B. $1 / \sqrt{\text{fan_in} + \text{fan_out}}$, where fan_in and fan_out are the number of input and output neurons of the layer
- C. The inverse of the number of training examples
- D. A fixed constant of 0.01 regardless of layer size

Q73. He initialisation is designed specifically for networks using which activation function?

- A. Sigmoid
- B. Tanh
- C. ReLU
- D. Linear

Q74. He initialisation scales initial weights by $\sqrt{2 / \text{fan_in}}$ rather than Xavier's $\sqrt{1 / \text{fan_in}}$ because:

- A. ReLU is twice as fast as sigmoid, requiring larger initial weights
- B. ReLU zeroes out negative pre-activations, effectively halving the variance of activations; the factor of 2 compensates for this
- C. He initialisation is designed for deeper networks where the fan_in is always larger than fan_out
- D. The factor of 2 accounts for the bias term which doubles the effective input dimension

Q75. Which initialisation strategy is most suitable for a network using sigmoid activations?

- A. He initialisation
- B. Zero initialisation
- C. Xavier (Glorot) initialisation
- D. Random uniform initialisation with a range of $(-10, 10)$

SECTION 3: BASICS OF IMAGE PROCESSING

- Q76. A greyscale image of size 28×28 pixels is represented as:
- A. A 3D array of shape (28, 28, 3)
 - B. A 2D array of shape (28, 28) where each element stores one intensity value
 - C. A 1D array of 784 RGB tuples
 - D. A 4D array including batch, height, width, and channel dimensions
- Q77. In an RGB image, each pixel is represented by:
- A. A single intensity value between 0 and 1
 - B. Three values corresponding to the red, green, and blue channel intensities
 - C. Four values including an alpha (transparency) channel
 - D. A binary value indicating foreground or background
- Q78. Pixel values in an 8-bit greyscale image range from 0 to 255, where 0 represents:
- A. White (maximum brightness)
 - B. Black (minimum brightness)
 - C. The median grey value
 - D. A transparent pixel
- Q79. Normalising pixel values by dividing by 255 is a common preprocessing step because:
- A. It converts the image from RGB to greyscale
 - B. It scales pixel values to the range [0, 1], which stabilises gradient descent and speeds up convergence
 - C. It removes the bias from all pixel values
 - D. It increases the effective resolution of the image
- Q80. Image data augmentation during training aims to:
- A. Reduce the resolution of images to speed up training
 - B. Artificially expand the training set by applying transformations, improving generalisation
 - C. Convert all training images to a uniform colour space
 - D. Encode images as dense numerical vectors before passing to the network
- Q81. A histogram of pixel intensities of an image is useful for:
- A. Computing the spatial frequency content of the image
 - B. Understanding the distribution of brightness values and guiding contrast enhancement
 - C. Detecting the edges and contours in the image
 - D. Identifying the dominant colour channel in the image
- Q82. Converting an RGB image to a greyscale image typically uses the formula:
- A. $\text{Grey} = (R + G + B) / 3$
 - B. $\text{Grey} = 0.299R + 0.587G + 0.114B$, weighted by human perceptual sensitivity
 - C. $\text{Grey} = \max(R, G, B)$
 - D. $\text{Grey} = R * G * B / 255^2$
- Q83. Image standardisation (zero-mean, unit variance) differs from normalisation (scaling to [0,1]) in that:
- A. Standardisation is only applicable to greyscale images
 - B. Standardisation centres the pixel distribution around zero and scales by standard deviation, while normalisation only scales the range
 - C. Normalisation always produces better results than standardisation for CNN training
 - D. Standardisation converts pixel values to integers while normalisation converts them to floats

- Q84. A convolution of an image with a Gaussian kernel is used to:
- A. Sharpen edges and increase image contrast
 - B. Blur the image and reduce high-frequency noise
 - C. Detect horizontal edges in the image
 - D. Convert the image from spatial domain to frequency domain
- Q85. The Sobel filter is primarily used in image processing for:
- A. Noise reduction by averaging neighbouring pixels
 - B. Edge detection by computing the gradient magnitude in horizontal and vertical directions
 - C. Contrast enhancement by stretching the pixel histogram
 - D. Morphological operations like dilation and erosion
- Q86. Applying a 3×3 mean (averaging) filter to an image replaces each pixel value with:
- A. The maximum pixel value in its 3×3 neighbourhood
 - B. The average of all pixels in its 3×3 neighbourhood
 - C. The minimum pixel value in its 3×3 neighbourhood
 - D. The original pixel value plus the sum of its neighbours
- Q87. The Fourier transform of an image converts it from the spatial domain to the frequency domain. High frequencies in an image correspond to:
- A. Smooth, slowly varying regions
 - B. Rapid intensity changes such as edges and fine textures
 - C. The overall brightness (DC component) of the image
 - D. The colour channels of the image
- Q88. Median filtering is preferred over mean filtering for images corrupted by salt-and-pepper noise because:
- A. It is computationally cheaper for large kernel sizes
 - B. It replaces each pixel with the median of its neighbourhood, which is robust to extreme outlier pixel values
 - C. It preserves high-frequency components better than mean filtering
 - D. It can be applied to colour images without any modification
- Q89. Histogram equalisation is applied to an image to:
- A. Reduce the image to greyscale
 - B. Redistribute pixel intensities to achieve a more uniform histogram, improving global contrast
 - C. Detect salient regions with the highest pixel intensity
 - D. Apply a gamma correction to correct for camera exposure
- Q90. Morphological dilation applied to a binary image has the effect of:
- A. Shrinking the foreground (white) regions
 - B. Expanding the foreground (white) regions and filling small holes
 - C. Detecting the edges of foreground regions
 - D. Converting the binary image to greyscale
- Q91. Resizing an image from 256×256 to 224×224 using bilinear interpolation:
- A. Crops the central 224×224 region of the original image
 - B. Estimates new pixel values by weighted averaging of the four nearest pixels in the original image
 - C. Discards every 32nd pixel to achieve the target resolution
 - D. Adds a border of black pixels to pad the image to the target size
- Q92. Horizontal flipping of an image during data augmentation is semantically valid for a digit recognition task ONLY if:
- A. All digits look the same when flipped (which they do not, e.g., 6 and 9 look different flipped)
 - B. The dataset is perfectly balanced across all digit classes
 - C. The image has been standardised before flipping
 - D. The CNN uses global average pooling instead of fully connected layers
- Q93. Image padding in convolutional networks is used to:

- A. Increase the pixel depth (number of channels) of the input image
- B. Control the spatial dimensions of the output feature map by adding rows/columns of zeros around the input
- C. Normalise the image pixel values before the first convolution
- D. Encode the image label into the spatial dimensions of the feature map

 SECTION 4: CONVOLUTIONAL NEURAL NETWORKS (CNN)

Q94. In a CNN, a convolutional filter (kernel) learns to detect:

- A. The global average of the entire input feature map
- B. Local spatial patterns such as edges, corners, and textures in its receptive field
- C. The class label associated with the entire input image
- D. The frequency spectrum of the input image

Q95. A 5×5 input feature map convolved with a 3×3 kernel using no padding and stride 1 produces an output of size:

- A. 5×5
- B. 4×4
- C. 3×3
- D. 2×2

Q96. The formula for computing the output spatial dimension after convolution is: $\text{output} = \text{floor}((\text{input} - \text{kernel} + 2 * \text{padding}) / \text{stride}) + 1$. For input=32, kernel=3, padding=1, stride=1, the output is:

- A. 30
- B. 31
- C. 32
- D. 33

Q97. 'Same' padding in a convolutional layer means:

- A. All filters in the layer have the same size
- B. The output spatial dimensions equal the input spatial dimensions (when stride=1)
- C. The padding is always equal to the kernel size
- D. The output feature map uses the same activation as the previous layer

Q98. Increasing the stride from 1 to 2 in a convolutional layer has the effect of:

- A. Doubling the number of parameters in that layer
- B. Reducing the spatial dimensions of the output feature map by approximately half
- C. Increasing the depth (number of channels) of the output feature map
- D. Removing the need for padding

Q99. A convolutional layer with 32 filters, each of size 3×3×3 (applied to an RGB image), contains how many trainable weight parameters (excluding biases)?

- A. $32 \times 3 \times 3 = 288$
- B. $32 \times 3 \times 3 \times 3 = 864$
- C. $3 \times 3 \times 3 = 27$
- D. $32 \times 3 = 96$

Q100. A key advantage of convolutional layers over fully connected layers for image processing is:

- A. Convolutional layers can process images of any arbitrary size
- B. Parameter sharing – the same filter weights are applied across all spatial locations, dramatically reducing the number of parameters
- C. Convolutional layers use non-linear activations while fully connected layers do not

D. Convolutional layers do not require backpropagation during training

Q101. Local connectivity in CNNs means that each neuron in a feature map:

- A. Is connected to every pixel in the input image
- B. Is connected only to a small local region (receptive field) of the previous layer
- C. Receives input only from neurons in the same channel
- D. Shares weights with all neurons in the output layer

Q102. Depthwise separable convolutions (used in MobileNet) improve efficiency over standard convolutions by:

- A. Replacing all convolutional layers with fully connected layers
- B. Factorising a standard convolution into a depthwise convolution (per channel) followed by a 1×1 pointwise convolution, significantly reducing computation
- C. Using binary weights instead of floating-point weights
- D. Applying convolutions only to every other spatial location

Q103. A 1×1 convolution in a CNN is used to:

- A. Detect point-level features with no spatial context
- B. Change the number of channels (depth) while preserving spatial dimensions, acting as a dimensionality reduction or expansion
- C. Replace max pooling to downsample the feature map
- D. Apply the softmax function to the feature map

Q104. Max pooling with a 2×2 window and stride 2 applied to a 4×4 feature map produces an output of size:

- A. 4×4
- B. 3×3
- C. 2×2
- D. 1×1

Q105. Max pooling contributes to translational invariance in CNNs because:

- A. It applies the same filter to all spatial positions
- B. Taking the maximum value in a region is insensitive to small shifts of the detected feature within that region
- C. It normalises the feature map to zero mean
- D. It converts the feature map from 2D to 1D

Q106. Global Average Pooling (GAP) at the end of a CNN architecture:

- A. Applies average pooling with a 2×2 window across the entire feature map
- B. Averages all spatial positions in each feature map channel to produce a single value per channel, replacing the fully connected classifier
- C. Averages the weights of all filters in the last convolutional layer
- D. Normalises each feature map to have the same average activation

Q107. Average pooling, compared to max pooling, tends to:

- A. Preserve stronger activations (edges) better
- B. Provide smoother, more diffuse feature maps by averaging all activations rather than taking the maximum
- C. Produce larger output feature maps
- D. Require more trainable parameters

Q108. Pooling layers have no trainable parameters. Their primary purpose is to:

- A. Apply learned non-linear transformations to feature maps
- B. Progressively reduce the spatial dimensions of feature maps, reducing computation and controlling overfitting
- C. Add batch normalisation to the feature maps
- D. Increase the number of feature channels at each layer

Q109. In a typical CNN for image classification, the depth of feature maps generally:

- A. Decreases from input to output, reducing computation

- B. Increases from input to output, as later layers detect more complex and abstract features
- C. Remains constant throughout the network
- D. Is equal to the number of training classes at every layer

Q110. Residual connections (skip connections) in ResNet address which training challenge?

- A. Overfitting caused by excessive data augmentation
- B. The degradation problem, where adding more layers causes training accuracy to saturate or decrease due to vanishing gradients
- C. The vanishing feature map problem caused by excessive pooling
- D. Overfitting caused by having too many filters in shallow layers

Q111. In a CNN, the receptive field of a neuron in a deep layer refers to:

- A. The number of filters applied to that neuron
- B. The region of the original input image that influences the neuron's activation
- C. The total number of parameters used to compute that neuron's activation
- D. The spatial size of the feature map at that layer

Q112. Why is flattening performed before the fully connected layers in a CNN classifier?

- A. To convert complex feature maps to a simpler greyscale representation
- B. To convert the 3D feature map tensor (height × width × channels) into a 1D vector that a Dense layer can process
- C. To apply global average pooling to the feature maps
- D. To normalise the feature map values before classification

Q113. Batch normalisation is commonly inserted in CNN architectures:

- A. Only before the first convolutional layer to normalise inputs
- B. After convolutional layers and before activation functions, to stabilise and accelerate training
- C. Only in the fully connected layers of the classifier head
- D. After the softmax layer to normalise class probabilities

Q114. The VGG architecture is characterised by:

- A. Very small networks with only 3 convolutional layers
- B. Using small 3×3 convolutional filters throughout, stacked to achieve large effective receptive fields with fewer parameters than large filters
- C. Using dilated convolutions with increasing dilation factors
- D. Replacing all pooling layers with strided convolutions

Q115. A Grad-CAM visualisation in a CNN helps to:

- A. Speed up the training of deep convolutional layers
- B. Highlight the spatial regions in the input image most responsible for a specific class prediction
- C. Initialise the convolutional filters using gradient information
- D. Select the optimal number of convolutional layers

SECTION 5: TRANSFER LEARNING

Q116. Transfer learning in deep learning refers to:

- A. Training a model from scratch on a new task using randomly initialised weights
- B. Reusing a model pre-trained on one task as the starting point for a different but related task
- C. Transferring the training data from one model to another
- D. Using the output of one model as the input features for a separate model without fine-tuning

Q117. Which of the following scenarios most justifies using transfer learning?

- A. A large labelled dataset is available for the target task and unlimited compute is available
- B. The target task has limited labelled data but a related large-scale pre-trained model exists
- C. The target task and source task have completely unrelated domains (e.g., medical images and audio spectrograms)
- D. The target task requires training from scratch to avoid negative transfer

Q118. In the 'feature extraction' approach to transfer learning, the pre-trained model's convolutional base is:

- A. Retrained completely on the new dataset
- B. Frozen (weights are not updated), and only a new classifier head is trained on top
- C. Replaced entirely by a new convolutional base with random weights
- D. Distilled into a smaller model before training on the new task

Q119. Fine-tuning a pre-trained model differs from feature extraction in that:

- A. Fine-tuning freezes the pre-trained layers; feature extraction does not
- B. Fine-tuning also updates some or all of the pre-trained model's weights during training on the new task
- C. Fine-tuning uses a higher learning rate than feature extraction
- D. Fine-tuning discards the pre-trained classifier head and retrain from scratch

Q120. When fine-tuning a pre-trained model, a lower learning rate is typically used for the pre-trained layers because:

- A. Pre-trained layers are closer to the output and receive larger gradients
- B. The pre-trained weights already contain useful representations; large updates would destroy this knowledge
- C. Lower learning rates cause faster convergence in fine-tuning compared to feature extraction
- D. The learning rate is automatically set by the pre-trained model's architecture

Q121. ImageNet pre-trained models (e.g., ResNet, VGG) are widely used as starting points for computer vision tasks because:

- A. ImageNet contains medical images that transfer well to all domains
- B. Training on 1.2 million images across 1,000 classes causes the model to learn rich, general visual features such as edges, textures, and object parts
- C. ImageNet pre-trained models have zero parameters in the final layer, making them adaptable to any task
- D. ImageNet models are trained without regularisation, making them more flexible for new tasks

Q122. Negative transfer in transfer learning occurs when:

- A. The model is too large for the available GPU memory
- B. Knowledge from the source domain hurts performance on the target domain, typically due to domain mismatch
- C. The pre-trained model's accuracy on the source task degrades after fine-tuning
- D. The learning rate is too high for the fine-tuning step

Q123. In transfer learning, which layers of a pre-trained CNN are most commonly frozen (kept unchanged) when the target dataset is small?

- A. Only the output layer
- B. All layers including the classifier head
- C. The earlier convolutional layers that capture general low-level features
- D. Only the batch normalisation layers

Q124. When using a pre-trained ImageNet model for a binary medical image classification task, the final layer of the original model (softmax over 1,000 classes) is typically:

- A. Kept unchanged and the model is used directly
- B. Replaced with a new output layer appropriate for binary classification (e.g., 1 neuron with sigmoid)
- C. Removed and the second-to-last layer's output is used directly as the classification decision
- D. Replaced with a 1,000-dimensional softmax and thresholded at 0.5

Q125. Domain adaptation is a form of transfer learning where the challenge is that:

- A. The target task has a different number of classes than the source task
- B. The source and target domains have the same task but different data distributions
- C. The target domain has no labelled data whatsoever
- D. The source model is too large to deploy on the target hardware

Q126. Progressive fine-tuning (unfreezing layers gradually from top to bottom) is beneficial because:

- A. It allows the model to learn the target task entirely from random weights
- B. It first adapts the task-specific layers to the new domain before allowing lower-level feature extractors to adjust, reducing the risk of catastrophic forgetting
- C. It automatically selects the optimal number of layers to freeze
- D. It eliminates the need for a separate validation set

Q127. Knowledge distillation is a transfer learning technique where:

- A. A smaller 'student' model is trained to mimic the output distribution of a larger 'teacher' model
- B. The weights of a large model are directly copied into a smaller model
- C. The source model is pruned by removing neurons with the smallest weights
- D. Multiple pre-trained models are averaged to create a combined model

SECTION 6: NATURAL LANGUAGE PROCESSING (NLP)

Q128. Tokenisation in NLP refers to:

- A. Converting text to numerical embeddings using a pre-trained model
- B. Splitting raw text into individual units (tokens) such as words, subwords, or characters
- C. Removing all punctuation and special characters from a text
- D. Converting all text to lowercase

Q129. Stop words in NLP are:

- A. Words that indicate the end of a sentence
- B. Common function words (e.g., 'the', 'is', 'at') that carry little semantic meaning and are often removed
- C. Out-of-vocabulary words that cannot be represented by the model
- D. Words that appear only once in the entire corpus

Q130. Stemming and lemmatisation both aim to reduce words to their root form, but they differ in that:

- A. Stemming requires a part-of-speech tagger while lemmatisation does not
- B. Stemming applies rule-based truncation (may produce non-words), while lemmatisation uses vocabulary and morphological analysis to produce valid dictionary forms
- C. Lemmatisation is faster but less accurate than stemming
- D. Stemming always produces longer word forms than lemmatisation

- Q131. Which of the following text preprocessing steps would be INAPPROPRIATE for a sentiment analysis task?
- A. Lowercasing all text
 - B. Removing negation words like 'not' and 'never'
 - C. Removing stop words
 - D. Removing HTML tags
- Q132. Subword tokenisation (e.g., Byte Pair Encoding used in GPT) addresses which NLP challenge?
- A. The need to convert text to greyscale for CNN processing
 - B. The out-of-vocabulary (OOV) problem, by representing rare or unknown words as sequences of common subword units
 - C. The requirement for large memory to store word-level vocabularies
 - D. The inability of word-level tokenisers to handle punctuation
- Q133. Named Entity Recognition (NER) is the NLP task of:
- A. Classifying the overall sentiment of a document as positive or negative
 - B. Identifying and classifying named entities (e.g., persons, organisations, locations) in text
 - C. Translating text from one language to another
 - D. Summarising a long document into a shorter version
- Q134. A Bag-of-Words (BoW) representation of a document ignores:
- A. The frequency of each word in the document
 - B. The order and position of words within the document
 - C. Whether a word is a stop word
 - D. The length of the document
- Q135. In the TF-IDF formula, the Inverse Document Frequency (IDF) component serves to:
- A. Increase the weight of words that appear frequently across many documents
 - B. Decrease the weight of words that appear in many documents (common words) and increase the weight of rare, informative words
 - C. Normalise the term frequency by the total number of words in the document
 - D. Convert word frequencies to probabilities
- Q136. The TF-IDF score for the word 'machine' in a document about machine learning, given that 'machine' appears in 80% of all documents in the corpus, will be:
- A. Very high, because 'machine' is central to the document's topic
 - B. Relatively low, because the high document frequency reduces the IDF component significantly
 - C. Exactly 1.0, because TF and IDF values are always between 0 and 1
 - D. Undefined, because the word 'machine' is a stop word
- Q137. A major limitation of both BoW and TF-IDF representations is:
- A. They are computationally too expensive for large vocabularies
 - B. They do not capture semantic relationships between words ('cat' and 'kitten' are treated as completely unrelated)
 - C. They can only be applied to documents with fewer than 1,000 words
 - D. They require labelled data to build the vocabulary
- Q138. N-gram language models extend BoW by:
- A. Weighting words by their position in the sentence
 - B. Including sequences of n consecutive words as features, capturing limited local word order
 - C. Using neural networks to compute continuous word representations
 - D. Replacing words with their part-of-speech tags before vectorisation
- Q139. Word2Vec learns word embeddings by training a shallow neural network to predict:
- A. The part-of-speech tag of each word in the sentence
 - B. A target word from its context words (CBOW) or context words from a target word (Skip-gram)

- C. The sentiment label of the sentence containing each word
- D. The character-level composition of each word

Q140. A classic property of word embeddings trained with Word2Vec is captured by the analogy:

- A. $\text{vector}(\text{'man'}) + \text{vector}(\text{'woman'}) = \text{vector}(\text{'human'})$
- B. $\text{vector}(\text{'king'}) - \text{vector}(\text{'man'}) + \text{vector}(\text{'woman'}) \approx \text{vector}(\text{'queen'})$
- C. $\text{vector}(\text{'cat'}) \times \text{vector}(\text{'dog'}) = \text{vector}(\text{'pet'})$
- D. $\text{vector}(\text{'Paris'}) / \text{vector}(\text{'France'}) = \text{vector}(\text{'capital'})$

Q141. GloVe (Global Vectors for Word Representation) differs from Word2Vec in that:

- A. GloVe uses recurrent neural networks while Word2Vec uses feedforward networks
- B. GloVe leverages global word co-occurrence statistics from the entire corpus, while Word2Vec uses local context windows
- C. GloVe produces character-level embeddings while Word2Vec produces word-level embeddings
- D. GloVe cannot be used for downstream NLP tasks without fine-tuning

Q142. FastText differs from Word2Vec by:

- A. Using attention mechanisms to compute context-sensitive word embeddings
- B. Representing each word as a bag of character n-grams, allowing it to generate embeddings for out-of-vocabulary words
- C. Using a bidirectional LSTM to compute embeddings from both left and right context
- D. Training exclusively on sentence pairs rather than individual words

Q143. Contextual word embeddings (e.g., from BERT or ELMo) differ from static embeddings (Word2Vec, GloVe) in that:

- A. Contextual embeddings have a fixed vector per word regardless of context
- B. Contextual embeddings produce different vectors for the same word depending on its context in the sentence
- C. Contextual embeddings are always lower-dimensional than static embeddings
- D. Contextual embeddings can only be computed for sentences shorter than 128 words

Q144. A Recurrent Neural Network (RNN) differs from a feedforward network in that:

- A. RNNs use convolutional filters to process sequential inputs
- B. RNNs maintain a hidden state that captures information from previous time steps, allowing them to process sequential data
- C. RNNs have no trainable parameters; all computations are fixed
- D. RNNs apply a softmax at every time step regardless of the task

Q145. The hidden state in an RNN at time step t is computed as:

- A. $h_t = \text{activation}(W_x * x_t)$
- B. $h_t = \text{activation}(W_h * h_{t-1} + W_x * x_t + b)$
- C. $h_t = h_{t-1} + x_t$
- D. $h_t = \text{softmax}(W_x * x_t + W_h * h_{t-1})$

Q146. The vanishing gradient problem is particularly severe in RNNs for long sequences because:

- A. RNNs apply the same weight matrix at every step, causing gradients to be multiplied by this matrix many times, often shrinking exponentially
- B. RNNs use tanh activations which have a gradient larger than 1 for all inputs
- C. RNNs cannot receive gradients from time steps more than 5 steps in the past
- D. The hidden state dimension is typically too small to carry gradient information

Q147. An LSTM (Long Short-Term Memory) network addresses the vanishing gradient problem using:

Q 1 B	Q 2 B	Q 3 C	Q 4 C	Q 5 B	Q 6 B	Q 7 C	Q 8 B	Q 9 B	Q 10 B
Q 11 B	Q 12 A	Q 13 B	Q 14 B	Q 15 B	Q 16 C	Q 17 B	Q 18 C	Q 19 C	Q 20 B
Q 21 B	Q 22 B	Q 23 B	Q 24 C	Q 25 B	Q 26 B	Q 27 B	Q 28 B	Q 29 B	Q 30 B
Q 31 B	Q 32 B	Q 33 B	Q 34 B	Q 35 B	Q 36 B	Q 37 B	Q 38 B	Q 39 B	Q 40 B
Q 41 B	Q 42 A	Q 43 B	Q 44 B	Q 45 B	Q 46 B	Q 47 B	Q 48 C	Q 49 B	Q 50 B
Q 51 B	Q 52 B	Q 53 B	Q 54 B	Q 55 B	Q 56 B	Q 57 B	Q 58 B	Q 59 B	Q 60 B
Q 61 B	Q 62 B	Q 63 B	Q 64 C	Q 65 B	Q 66 B	Q 67 B	Q 68 B	Q 69 B	Q 70 B
Q 71 B	Q 72 B	Q 73 C	Q 74 B	Q 75 C	Q 76 B	Q 77 B	Q 78 B	Q 79 B	Q 80 B
Q 81 B	Q 82 B	Q 83 B	Q 84 B	Q 85 B	Q 86 B	Q 87 B	Q 88 B	Q 89 B	Q 90 B

Q 91 B	Q 92 A	Q 93 B	Q 94 B	Q 95 C	Q 96 C	Q 97 B	Q 98 B	Q 99 B	Q100 B
Q101 B	Q102 B	Q103 B	Q104 C	Q105 B	Q106 B	Q107 B	Q108 B	Q109 B	Q110 B
Q111 B	Q112 B	Q113 B	Q114 B	Q115 B	Q116 B	Q117 B	Q118 B	Q119 B	Q120 B
Q121 B	Q122 B	Q123 C	Q124 B	Q125 B	Q126 B	Q127 A	Q128 B	Q129 B	Q130 B
Q131 B	Q132 B	Q133 B	Q134 B	Q135 B	Q136 B	Q137 B	Q138 B	Q139 B	Q140 B
Q141 B	Q142 B	Q143 B	Q144 B	Q145 B	Q146 A	Q147 B	Q148 B	Q149 B	Q150 B